

Student number: _____
2022 Preliminary Higher School Certificate Examination

Mathematics Standard

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen.
- NESA-approved calculators may be used
- A reference sheet is provided at the back of this paper

For questions in Section II show relevant mathematical reasoning and/or calculations

Total marks -100

Section I - 15 marks (pages 2 - 7)

- Attempt Questions 1 – 15
- Allow about 20 minutes for this section

Section II – 85 marks (pages 8 – 28)

- Attempt Questions 16 – 35
- Allow about 1 hour and 40 minutes for this section

Care has been taken to ensure that this paper is free of errors and that it follows the spirit of the syllabus. Some questions have been adapted from previous Preliminary examinations and various sources, in an attempt to provide students with exposure to a broad range of questions.

Section I

15 marks

Attempt questions 1 - 15

Allow about 20 minutes for this section

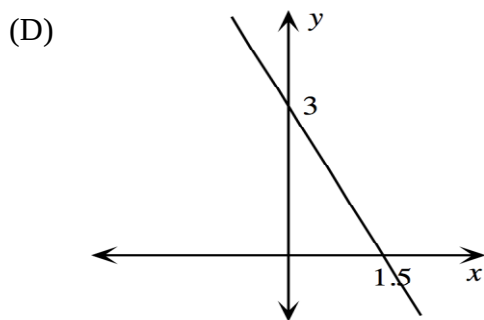
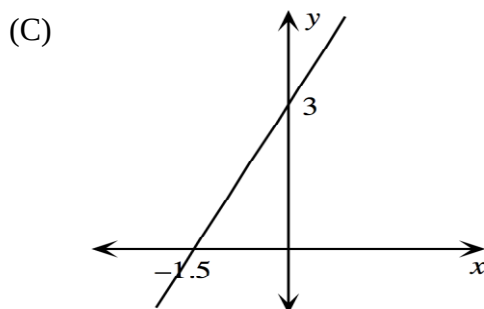
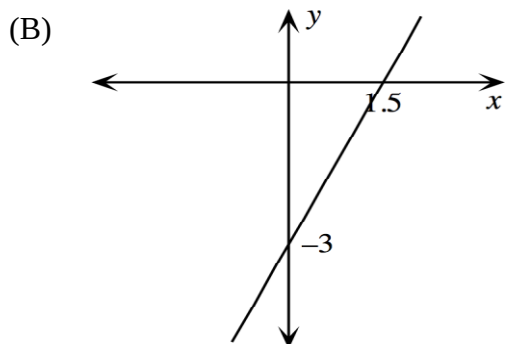
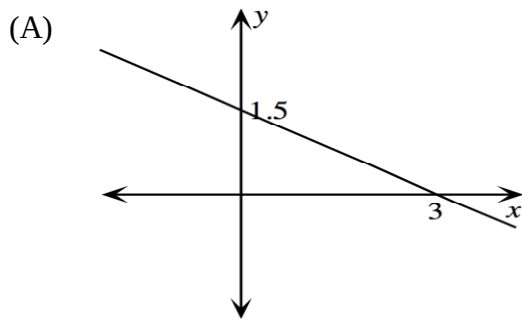
Use the multiple-choice answer sheet for questions 1-15

1. The formula for converting Fahrenheit temperatures (F) to Celsius (C) is

$$C = \frac{5}{9}(F - 32). \text{ Convert } 113^{\circ} \text{ Fahrenheit to Celsius degrees.}$$

- (A) 22°
 - (B) 34°
 - (C) 45°
 - (D) 68°
2. Sally earns a retainer of \$516 per week plus commission of 6.5% on the value of books sold. If Sally earned \$1950 in a week, what was the value of the books sold that week?
- (A) \$11 002.34
 - (B) \$13 145.26
 - (C) \$16 270.80
 - (D) \$22 061.54
3. Convert 8 m^3 to cm^3
- (A) $8\,000 \text{ cm}^3$
 - (B) $8\,000\,000 \text{ cm}^3$
 - (C) $80\,000\,000 \text{ cm}^3$
 - (D) $800\,000\,000 \text{ cm}^3$

4. Which of the following could be the graph of $y = 2x - 3$?



5. An investment of \$20 000 earns interest at 12% p.a., compounding half-yearly. Which of the following expressions gives the value of this investment after 6 years?

- (A) $\$20\,000 \times 1.06^{12}$
- (B) $\$20\,000 \times 0.06^{12}$
- (C) $\$20\,000 \times 0.12 \times 6$
- (D) $\$20\,000 \times 0.06 \times 12$

6. An archer has a 60% chance of scoring a bulls-eye with each arrow he fires. What is the probability that he scores a bulls-eye with two consecutive arrows?

(A) 22%

(B) 30%

(C) 36%

(D) 60%

7. Samuel has a reaction time of 1.8 seconds, he was travelling at 70km/h when he saw a large tree had fallen across the road. He applied his brakes and stopped 45m later. Calculate his stopping distance, correct to 1 decimal place.

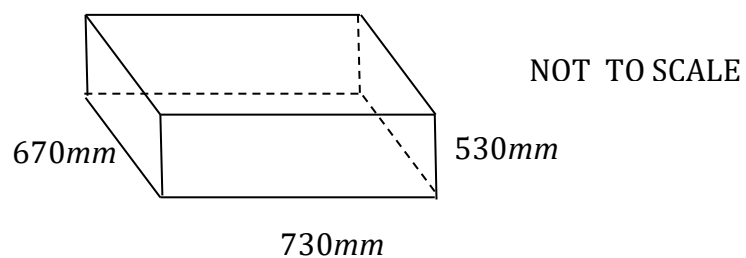
(A) 45 m

(B) 60.1 m

(C) 80.0 m

(D) 130 m

8. The surface area, in square metres, of the following closed rectangular prism is?



(A) 0.259223 m^2

(B) 1.345 m^2

(C) 1.6884 m^2

(D) 2.4622 m^2

9. Which of the following data would be classified as discrete?

- (A) The number of people at a vaccine clinic.
- (B) The speed of a plane.
- (C) The weight of a woman.
- (D) The height of a male football player.

10. The table below gives the longitudes of 3 world cities.

Midway Island	London	Sri Lanka
-11 hrs (UTC)	0°	+5.5 hrs (UTC)

It is 3a.m. in London on a Tuesday.

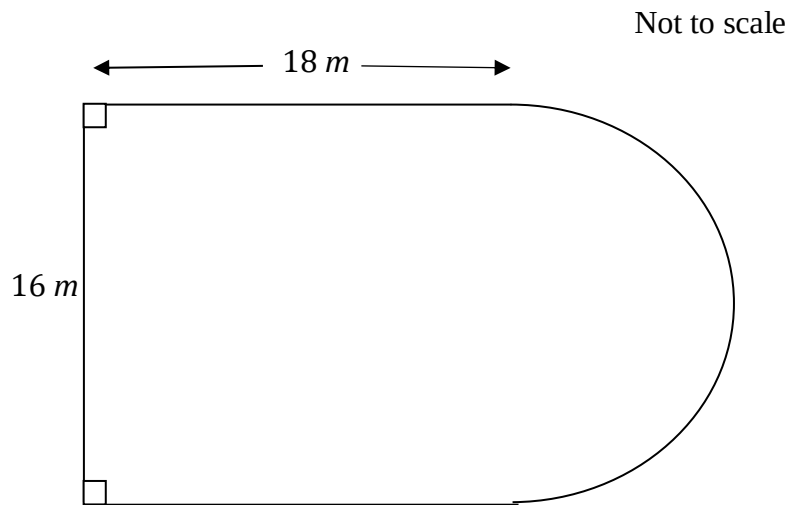
Which of the following statements is **True** ?

- (A) There is a 5.5 hr hour time difference between Midway Island and Sri Lanka
- (B) It is 8:30 am Monday in Sri Lanka
- (C) It is 4:00 pm Sunday in Midway Island
- (D) There is a 16.5 hour time difference between Midway Island and Sri Lanka

11. Nina left Sydney at 10 a.m. to travel a distance of 1 000 km to Melbourne. She arrived in Melbourne at 9.30 p.m. What was her average speed for the journey, correct to the nearest km/h?

- (A) 60 km/h
- (B) 82 km/h
- (C) 87 km/h
- (D) 100 km/h

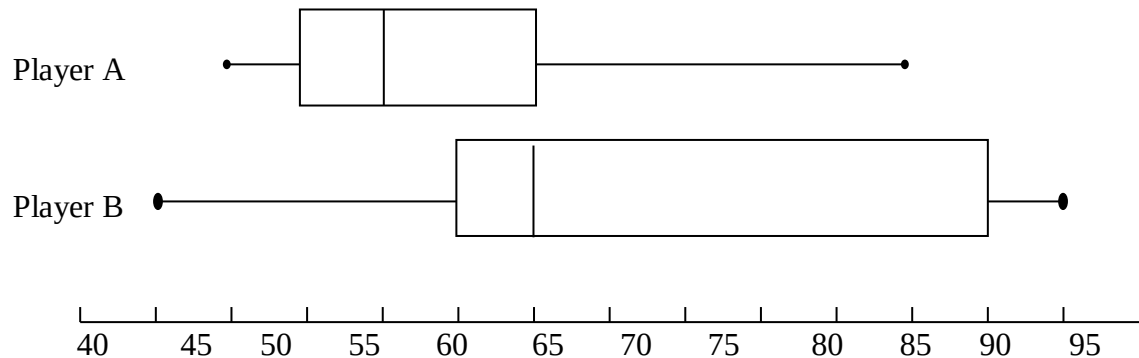
12. The diagram shows the surface area of a pool, which consists of a rectangle and a semicircle. The pool has vertical sides and a constant depth of 1.9m.



Which expression shows the maximum volume of water, in litres, the pool can hold.

- (A) $(16 \times 18 \times \pi \times 8^2) \times 1.9 \times 1000$
- (B) $(16 \times 18 + \pi \times 8^2) \times 1.9 \times 1000$
- (C) $\left(16 \times 18 + \frac{1}{2} \times \pi \times 8^2\right) \times 1.9$
- (D) $\left(16 \times 18 + \frac{1}{2} \times \pi \times 8^2\right) \times 1.9 \times 1000$
13. A car uses 8 litres of petrol to travel 128 km. How far can the car travel on 65.5 litres of petrol?
- (A) 4 km
- (B) 1048 km
- (C) 1456 km.
- (D) 1567 km

14. The box and whisker plots shown below compare the golf scores of two players over 9 holes. Which of the following statements is correct?



Which of the following is correct?

- (A) Player A is negatively skewed
 - (B) The range of Player B is smallest.
 - (C) Player A is positively skewed
 - (D) The median of Player A is greatest.
15. In the half yearly examination, Barry sat for five papers and the mean of his marks was 55 with a standard deviation of 12. In the yearly examination he improved by 15 marks on every paper, compared to the half yearly. What would be the effect of this on the mean and standard deviation of his yearly marks compared to the half yearly? .
- (A) The mean and standard deviation will both increase.
 - (B) The mean will increase and the standard deviation will be the same.
 - (C) The mean will be the same and the standard deviation will increase.
 - (D) The mean and standard deviation will both be the same.

Section II

85 marks

Attempt all questions

Allow about 1 hour and 40 minutes for this section

Answer the questions in the spaces provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the back of the examination paper.

Question 16 (6 marks)

Marks

An offset survey of a building site was conducted. The notebook entries are shown.

Measurements are in metres.

	D	
	80	
B 35	65	
	40	45 C
	0	
	A	

- (a) Use these notebook entries to draw a neat sketch of the building site. 2
Mark the interval and offset measurements on your diagram.

- (b) What is the perimeter of the building site, to the nearest metre? 2

(c) Calculate the area of the site.

2

Question 17 (3 marks)

Marks

If your taxable income is less than or equal to \$19 500, your Medicare levy is nil.
If your taxable income is:

- more than \$19 500 but less than \$28 481, your Medicare levy is 26 cents for every dollar above \$19 500;
- more than or equal to \$28 481, your Medicare levy is 4.75% of your taxable income.

(a) Calculate the Medicare levy, if Kim has a taxable income of \$23 490

1

(b) If Liong has a taxable income of \$38 438. Calculate how much more Liong's Medicare levy will be compared to Kim's Medicare levy?

2

Question 18

(4 marks)

Marks

David is an 18 year old school boy who has a part time job at a local department store.

He works on Thursday nights from 4 pm to 9 pm and on Saturday from 8:30 am to 4:30 pm. He is paid an hourly rate of \$15.45. However, on Saturday he is paid 'time and a half' for any work after 12 noon.

- (a) Calculate David 's weekly wage.

1

- (b) David wishes to buy a used car which cost \$2400. Other expenses which he needs to consider are:

1

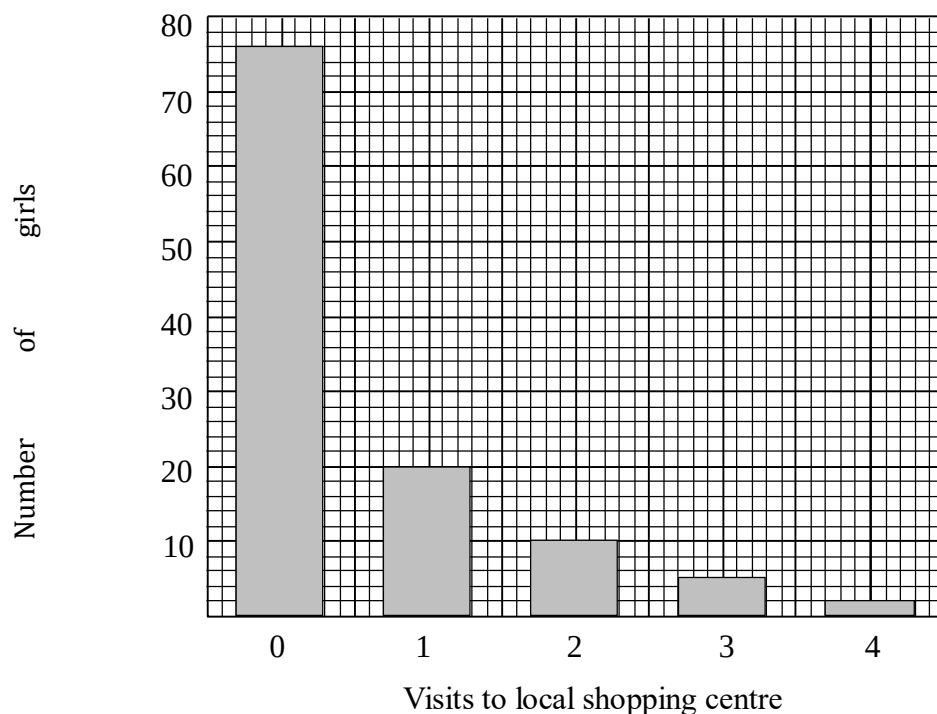
Car Insurance	\$935
Stamp Duty	\$48
Road Service Fee	\$75

Find the total cost of the car plus expenses.

- (c) David saves 40% of his weekly wage.
Find how many weeks he will have to work to save enough money to buy the car plus expenses. (Answer to the nearest whole week.)

2

A group of 16 year old girls were surveyed. The graph shows the number of times they visited their local shopping centre during a week.



- (a) How many girls were surveyed?

1

- (b) Find the probability that at least one girl visited the shopping centre during the week.

2

(a)

After a recent drought the rain gauge on a farm registered 84 mm of rain during a storm. A barn on the farm has a rectangular roof measuring 34 metres by 35 metres.
How many kilolitres of water fell on barn roof during the storm? Answer to nearest kL

2

(b)

The rain falling on the barn roof is collected in a cylindrical tank with a diameter of 3 metres. The tank was empty before the storm.
Find the depth of water in the tank after the storm.
Answer correct to two decimal places.

2

(c)

The height of the tank is 18 metres. How many more kilolitres can fall before the tank begins to overflow? Answer correct to two decimal places.

2

Question 21 (2 marks)

Marks

The length of the cross-sectional diagonal (C) of an object is given by: $C = \sqrt{5s^2}$ **2**
where s is the length of the side of the object.

Write down an equation which gives the length of the side of the object.

Questions 16 to 21 are worth 24 marks

Question 22 (2 marks)

Marks

Expand and simplify $p(p - q) - q(q - p)$

2

Question 23 (3 marks)

Marks

Young's rule can be used to calculate a child's medicine dose. Young's rule is:
 $C = \frac{nA}{n + 12}$ where C is the child's dose (in mL), n is the age of the child (in years)
and A is the adult dose (in mL).

For a particular medicine, the adult dose is 28 mL.

(a) What is the dose for a 5 year old child?

1

(b) Calculate the difference in dosage between a 5 year old and a 9 year old child.

2

Brad is a doctor who has checked the pulse rate of his male and female patients. The table below shows his results.

Male pulse rate	64	59	93	84	63	56	80	76	91	98
Female pulse rate	52	84	77	94	61	99	73	68	77	86

(a)

Display the information above in an **ordered** back-to-back stem-and-leaf plot.

2

(b)

Calculate the mean and standard deviation for both male and female patients.

2

(c)

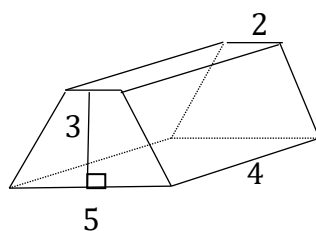
Find the five-number summary for both male and female patients.

3

Question 25 (7 marks)

Marks

A hollow trapezoidal metal wedge is being used at a building site. All dimensions are in metres.



- (a) Find the volume of the container? (Answer to nearest cubic metre)

1

- (b) The hollow container is filled with crude oil. Find how many kilolitres of crude oil is needed to fill 20 wedges?

1

- (c) The container has a length of $4m$. Calculate its absolute error and state the limits of accuracy.

2

- (d) Find the percentage error for the length and height of the container. 2

- (e) Which of the two measurements is more accurate? 1

Question 26 (4 marks)

Marks

The charges for gas over a 3-month period are shown below.

Usage charge	First 4000 MJ	\$0.07680 per MJ
	Additional MJ over 4000	\$0.03220 per MJ

- (a) Nina used 7000 MJ of gas in this period. What is the charge? 1

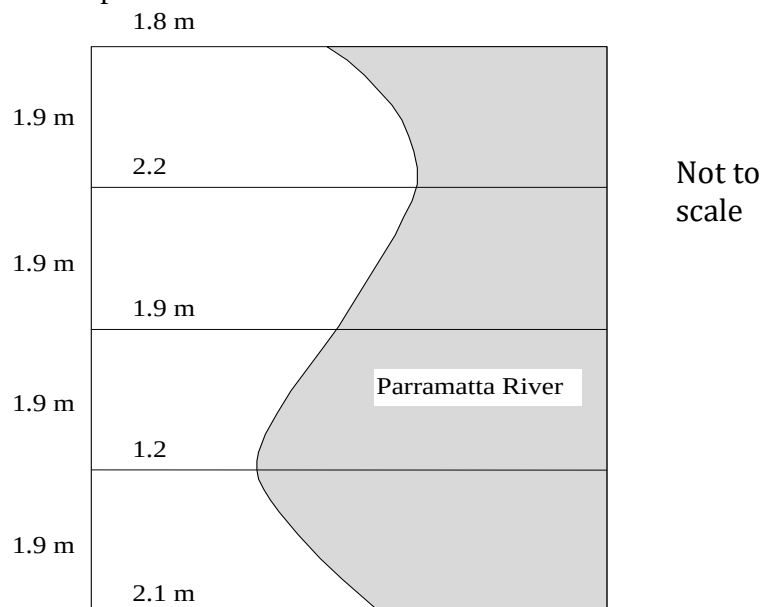
- (b) Nina, owns a large shop and has decided to increase her energy bills. She decided to increase her gas usage by 2.5%, during the next financial year. Find the cost of her gas bill during the next financial year..

3

Question 27 (4 marks)

Marks

The diagram shows a fenced off backyard adjacent to the Parramatta River (shaded) and area to be paved.



- (a) Calculate the area to be paved using four applications of the Trapezoidal rule. Answer to one decimal place.

2

- (b) Paving tiles come in boxes of 25. Each individual tile has dimensions 30 cm x 15 cm and each box costs \$76.65. Calculate the cost of paving this area. 2

Question 28 (5 marks) **Marks**

The probability of rain falling on a particular night in September at a health resort is 75%.

Dino travels to the health resort for three nights.

- (a) Draw a tree diagram to show the outcomes of rain falling or not falling for the three days. 2

- (b) What is the probability of rain falling on all three nights? 1

- (c) Find the probability that there is rain falling on at least one night.

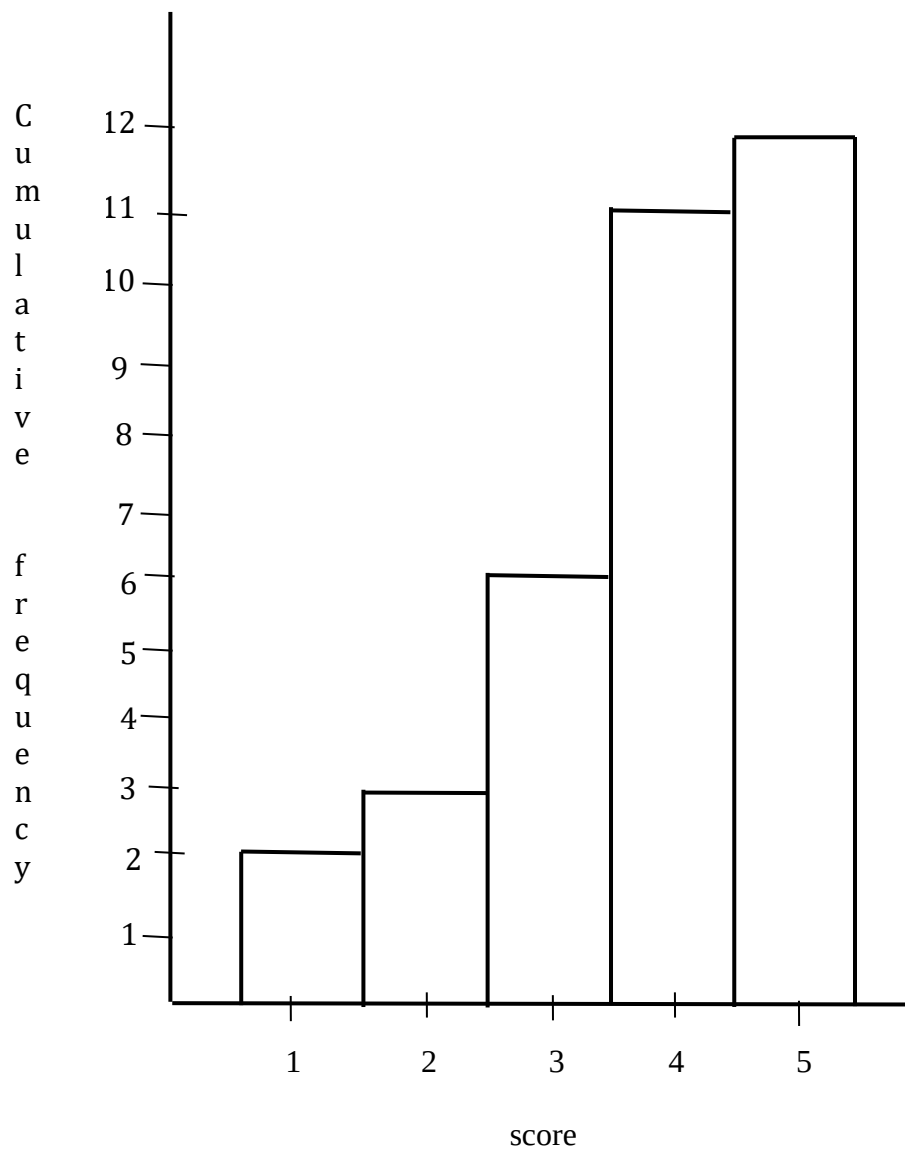
2

Questions 16 to 28 are worth 56 marks

Question 29 (4 Marks)

Marks

Mr Guass's Year 11 mathematics class completed a mathematics test.
The cumulative frequency histogram for the test is shown below



(a) Draw a frequency histogram for the mathematics test.

3

(b) Determine the mean mark for the mathematics test?

1

Question 30 (6 marks)

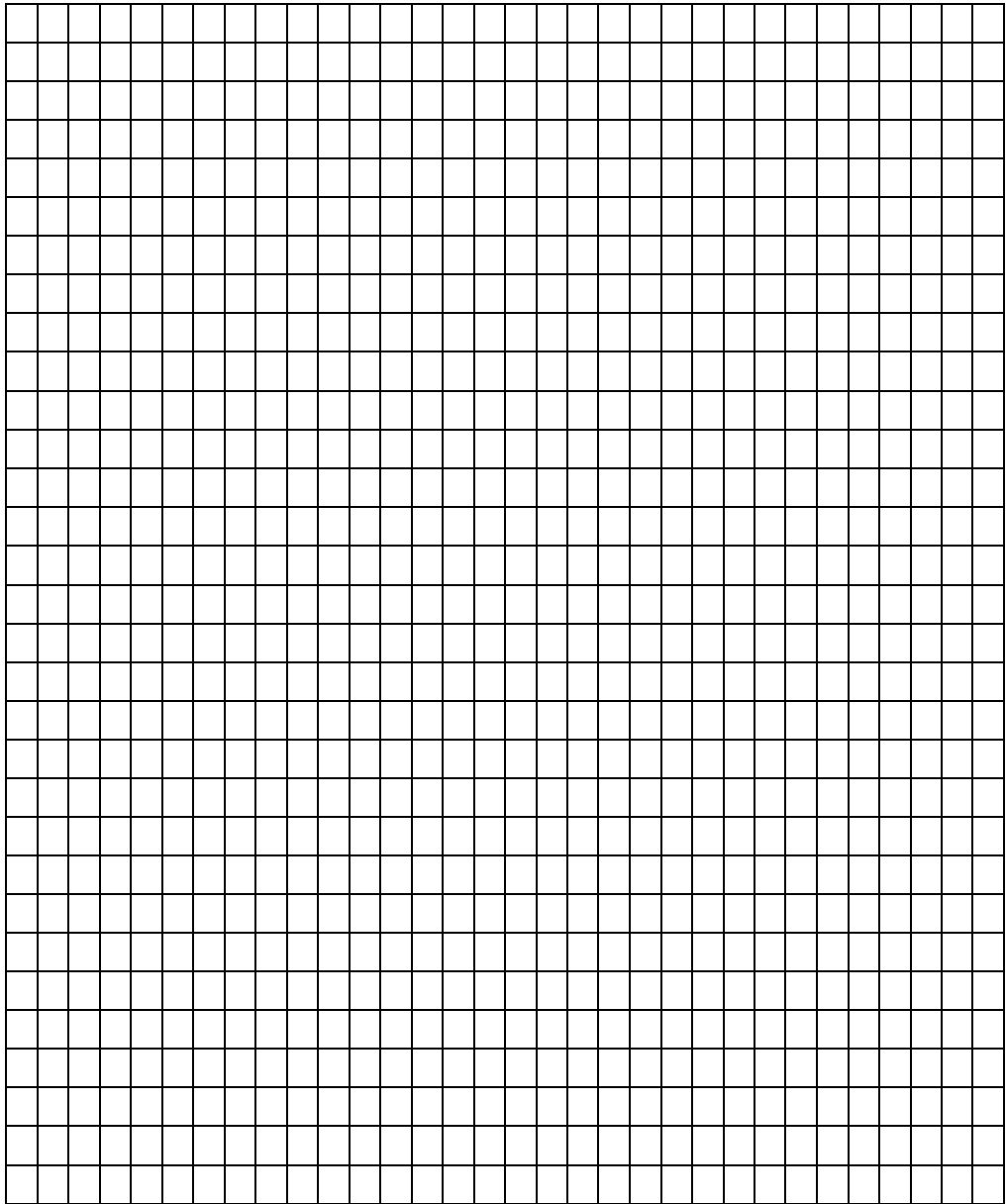
Marks

The table shows the linear relationship between distances measured in miles and distances measured in kilometres

Miles M	0	5	10	15	20	25	30	
Kilometres K	0	8	16	24	32	40	48	

(a) Graph this linear relationship on the number plane

2



(b) Find the equation of this line.

1

.....

.....

(c) What is the gradient of the line and what does it represent? 2

(d) Convert 18 miles to kilometres. 1

Question 31 (4 marks) **Marks**

The average rate of alcohol metabolism (breakdown) reduces a person's BAC by 0.019 units per hour. Leila has just measured his BAC and it is 0.193.

(a) How many hours and minutes must elapse before she is below the 0.05 limit and can legally drive again? 2

(b) How long must pass for her BAC to reach zero? 2

Question 32 (4 marks)

Marks

Vonnie is travelling from Sydney(UTC +10) to Iran(UTC +3.5) and then onto El Salvador (UTC -6).

The flight from Sydney to Iran takes 9 hours and 15 minutes. She stayed in Iran for 22 hours and 4 minutes. The flight from Iran to El Salvador takes 11 hours and 30 mins.

- (a) The plane left Sydney on Saturday at 10:00 am. What time is this in EL Salvador? **2**

- (b) Vonnie's mother wishes to meet her daughter when the plane lands in El Salvador. What is the time in El Salvador when the plane arrives? **2**

Question 33

(3 marks)

Marks

Compounded values of \$1		
	Interest rate per period	
Period	2%	5%
1	1.0200	1.0500
2	1.0404	1.1025
3	1.0612	1.1576
4	1.0824	1.2155
5	1.1041	1.2763
6	1.1262	1.3401

- (a) Use the table to calculate the final balance if \$700 is invested for 2 years at 5% p.a. **1**

- (b) Find the compound interest if \$600 is invested at 4% p.a. for 3 years compounded half yearly. **2**

(6 marks)

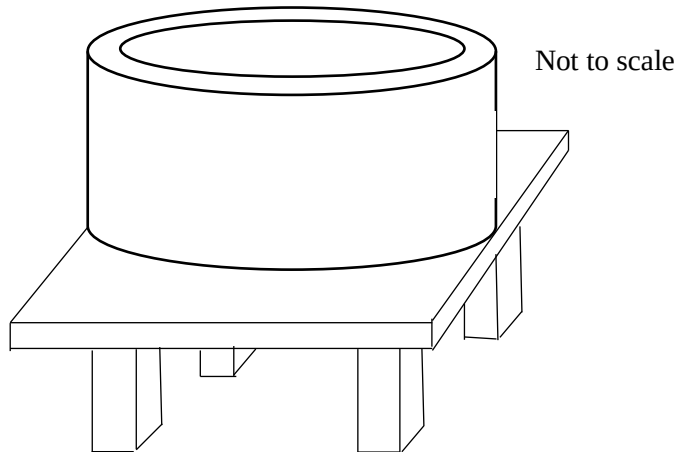
Marks

The following furniture item consists of an open cylindral container with thickness of 3 cm attached to a table. The legs of the table are all the same size.

The container is 80 cm high and external diameter of 70 cm.

The table has dimensions 70 cm x 90 cm x 12 cm

The width and depth of one leg of the table is 7 cm x 7 cm with height of 50 cm



(a) Calculate the total surface area of the above furniture item.

4

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

- (b) The Chan family run a furniture store and they order 300 of the above furniture items. Each furniture item is not varnished. 2
 A 250 ml tin of varnish costs \$29.50. Find the cost of varnishing all the furniture, if 250 ml of varnish covers 60 square metres.
 (Use $1m^2 = 10000\text{ cm}^2$)

Question 35 (2 Marks)

Income	Tax Payable
\$1 to \$6 000	\$0
\$6 001 to \$20 000	\$0 plus 17c for each \$1 over \$6 000
\$20 001 to \$50 000	\$2 380 plus 30c for each \$1 over \$20 000
\$50 001 to \$60 000	\$11 380 plus 42c for each \$1 over \$50 000
In excess of \$60 000	\$15 580 plus 47c for each \$1 over \$60 000

Using the tax scale find the taxable income if the tax paid is \$37 630.

End of Exam

Mathematics Standard 1 and Standard 2 Reference sheet

Measurement

Limits of accuracy

$$\text{Absolute error} = \frac{1}{2} \times \text{precision}$$

Upper bound = measurement + absolute error

Lower bound = measurement – absolute error

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \pi r^2$$

$$A = \frac{h}{2} (a + b)$$

$$A \approx \frac{h}{2} (d_f + d_l)$$

Surface area

$$A = 2\pi r^2 + 2\pi r h$$

$$A = 4\pi r^2$$

Trigonometry

$\sin A = \text{opp/hyp}$, $\cos A = \text{adj/hyp}$, $\tan A = \text{opp/adj}$

$$A = \frac{1}{2} a b \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Volume

$$V = \frac{1}{3} Ah$$

$$V = \frac{4}{3} \pi r^3$$

Financial Mathematics

$$FV = PV(1 + R)^n$$

Straight-line method for depreciation

$$S = V_0 - Dn$$

Declining-balance formula for depreciation

$$S = V_0(1 - r)^n$$

Statistical Analysis

An outlier is a score

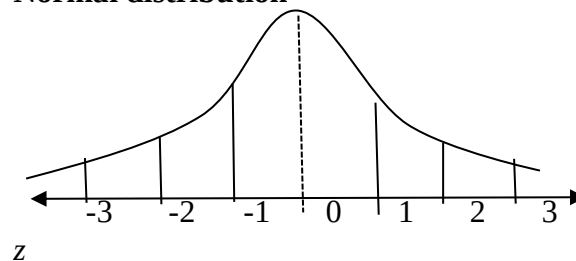
less than $Q_1 - 1.5 \times IQR$

or

more than $Q_3 + 1.5 \times IQR$

$$z = \frac{x - \bar{x}}{s}$$

Normal distribution



- Approximately 68% of scores have z-scores between -1 and 1
- Approximately 95% of scores have z-scores between -2 and 2
- Approximately 99.7% of scores have z-scores between -3 and 3

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Centre Number

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Student Number

YEAR 11**MATHEMATICS STANDARD -MULTIPLE CHOICE ANSWER SHEET**

Sample $2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9
 (A) ○ (B) ● (C) ○ (D) ○

(A) (B) (C) (D)

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A)  (B)  (C)  (D) 

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows

(A)  (B)  (C)  (D) 

correct 

correct

- 1** (A)○ (B)○ (C)○ (D)○

- 9 (A) ☐ (B) ☐ (C) ☐ (D) ☐

- 2 (A) ○ (B) ○ (C) ○ (D) ○

- 10** (A) ○ (B) ○ (C) ○ (D) ○

- 3** (A) ○ (B) ○ (C) ○ (D) ○

- 11** (A)○ (B)○ (C)○ (D)○

- 4 (A)○ (B)○ (C)○ (D)○

- 12 (A) ☐ (B) ☐ (C) ☐ (D) ☐

- 5 (A) ○ (B) ○ (C) ○ (D) ○

- 13** (A) ☐ (B) ☐ (C) ☐ (D) ☐

- 6 (A) ☐ (B) ☐ (C) ☐ (D) ☐

- 14 (A) ☐ (B) ☐ (C) ☐ (D) ☐

- 7 (A) ☐ (B) ☐ (C) ☐ (D) ☐

- 15 (A) ☐ (B) ☐ (C) ☐ (D) ☐

- 8 (A) ☐ (B) ☐ (C) ☐ (D) ☐